ExpenseMate - Software Requirements Specification

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1. Introduction

1.1 Purpose

This document specifies the requirements of ExpenseMate, a personal expense manager that lets users record income and expenses, set monthly budgets and view spending analytics.

1.2 Scope

ExpenseMate is a client-server system made of a Flutter client (Android, iOS, web), a Node.js REST API and a MySQL database. Cloud backup and CSV import are outside version 1.0 and are listed under future scope.

1.3 Definitions

Term	Meaning
OTP	One-time 6-digit code sent by email for verification or password reset
JWT	Signed token that authenticates each API request
Budget	Monthly spending limit set for one expense category
Account	A payment source: cash, bank, card or wallet

2. Overall Description

Item	Description
User	One user type: a registered individual who manages personal finances
Client environment	Android, iOS or a web browser running the Flutter app
Server environment	Node.js 18 or later with MySQL 8.0 or later; SMTP service for email
Constraints	Client and server must reach each other over a network; every protected request needs a valid JWT; amounts must be greater than zero
Assumptions	The user owns a working email address; each user works in one currency at a time

3. Functional Requirements

ID	Requirement
FR-01	The system shall let a user register with name, email, password, country and currency.
FR-02	The system shall email a 6-digit OTP that expires after 10 minutes and allows at most 5 attempts.
FR-03	The system shall allow the OTP to be resent, with a 60-second wait between requests.
FR-04	The system shall log in verified users with email and password and refuse unverified users.
FR-05	The system shall let a user reset a forgotten password using an emailed OTP.
FR-06	The system shall record income and expense transactions with title, amount, category, account, date, optional note and tags.

ID	Requirement
FR-07	The system shall let a user attach a receipt image from the camera or gallery to a transaction.
FR-08	The system shall list transactions with search, filters by type, category, account and date, and pagination.
FR-09	The system shall let a user delete a transaction.
FR-10	The system shall let a user create, edit and delete accounts and shall show each balance as opening balance plus income minus expenses.
FR-11	The system shall provide 13 default categories and let a user create, edit and delete categories with an icon and colour.
FR-12	The system shall let a user set a monthly budget for each expense category; a limit applies to later months until replaced.
FR-13	The system shall show budget progress as green below 85 percent, amber from 85 to 100 percent and red above 100 percent of the limit.
FR-14	The system shall show a dashboard with total balance, monthly income and expense, and the 10 most recent transactions.
FR-15	The system shall show analytics by day, week, month and year with trend charts and a category-wise breakdown.
FR-16	The system shall export transactions as CSV or PDF and email the file to the user.
FR-17	The system shall let a user edit name, email and avatar.
FR-18	The system shall let a user change the password after confirming the current one.

ID	Requirement
FR-19	The system shall let a user change currency and optionally convert all stored amounts by a given rate.
FR-20	The system shall let a user choose a primary colour and dark mode.
FR-21	The system shall cache user data on the device and treat it as fresh for 15 minutes.
FR-22	The system shall clear the session token and local cache on logout.

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Passwords shall be stored as bcrypt hashes and OTPs as SHA-256 hashes.
NFR-02	Security	All endpoints except health and authentication shall require a valid JWT; tokens expire after 7 days.
NFR-03	Security	Responses shall never include password hashes, and password reset shall not reveal whether an email exists.
NFR-04	Performance	Cached data shall be shown immediately on launch; list requests shall respond within 2 seconds on a local server.
NFR-05	Usability	The interface shall follow Material 3 with a consistent, user-selectable theme across all 22 screens.
NFR-06	Portability	The client shall run on Android, iOS and web from one code base.
NFR-07	Reliability	Currency conversion shall run as a single database transaction so it either fully succeeds or changes nothing.

ID	Category	Requirement
NFR-08	Data integrity	The database shall enforce foreign keys, unique category and account names per user, and amount checks.
NFR-09	Maintainability	The backend shall be layered (routes, controllers, services, utilities) and validate all input in one validator class.
NFR-10	Error handling	Errors shall use one JSON format with a message and a code.

5. Use Cases

The actor is the User; the SMTP email server is a secondary actor for OTP and export emails.

Use case	Related requirements
Register and verify email	FR-01, FR-02, FR-03
Log in, log out	FR-04, FR-22
Reset password	FR-05
Add, view and delete transactions	FR-06, FR-07, FR-08, FR-09
Manage accounts and categories	FR-10, FR-11
Set budget and view alerts	FR-12, FR-13
View dashboard and analytics	FR-14, FR-15
Export transactions	FR-16
Manage profile, currency and theme	FR-17, FR-18, FR-19, FR-20

6. External Interfaces

Interface	Description
User interface	22 Flutter screens with a bottom navigation bar for Home, Analytics, Add, Budget and Profile
Client to server	REST over HTTP using JSON, port 5000, Bearer JWT
Server to database	MySQL connection pool, port 3306
Server to email	SMTP, port 587

7. Future Scope

Cloud backup, CSV import, and wider use of biometric sign-in are planned for later versions.